

Inbetriebnahme BMP280

Scope

Inbetriebnahme mit gängigen Libraries liefert unterschiedliche Ausgaben, wie Gerät nicht angeschlossen (Adafruit) oder hang-ups (Grove).

Gerätetyp identifizieren

Es gibt unterschiedliche Typen:

BMP180: relativ groß, Druckmessung auf 0,2 hPa genau

BMP280: kompakte für mobile Geräte, Druckmessung auf 0,001 hPa genau

BME280: wie BMP280 aber zusätzlich mit Feuchtemessung

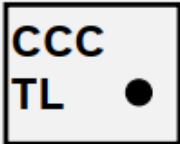
Gemäß Datenblatt von Bosch <https://www.bosch-sensortec.com/media/boschsensortec/downloads/datasheets/bst-bmp280-ds001.pdf>, S. 41, lässt sich der Sensortyp anhand des Codes auf dem Gehäuse identifizieren:

 BOSCH	Datasheet BMP280 Digital Pressure Sensor	Page 41
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7.3 Marking

7.3.1 Mass production devices

Table 30: Marking of mass production samples

Labeling	Name	Symbol	Remark
	Lot counter	CCC	3 alphanumeric digits, variable to generate mass production trace-code
	Product number	T	1 alphanumeric digit, fixed to identify product type, T = "K" "K" is associated with the product BMP280
	Sub-con ID	L	1 alphanumeric digit, variable to identify subcon (L = "S" for underfill 0273300436)
	Orientation marker	●	Vent hole

Hier:

9E8 – KS – BMP280 aus Massenproduktion



I2C-ID verifizieren

Datenblatt Bosch, S. 28:

The 7-bit device address is 111011x. The 6 MSB bits are fixed. The last bit is changeable by SDO value and can be changed during operation. Connecting SDO to GND results in slave address 1110110 (0x76); connection it to V_{DDIO} results in slave address 1110111 (0x77), which is the same

as BMP180's I²C address. The SDO pin cannot be left floating; if left floating, the I²C address will be undefined.

The I²C interface uses the following pins:

- SCK: serial clock (SCL)
- SDI: data (SDA)
- SDO: Slave address LSB (GND = '0', V_{DDIO} = '1')

CSB must be connected to V_{DDIO} to select I²C interface. SDI is bi-directional with open drain to GND: it must be externally connected to V_{DDIO} via a pull up resistor. Refer to chapter 6 for connection instructions.

The following abbreviations will be used in the I²C protocol figures:

- S Start
- P Stop
- ACKS Acknowledge by slave
- ACKM Acknowledge by master
- NACKM Not acknowledge by master

5.2.1 I²C write

Writing is done by sending the slave address in write mode (RW = '0'), resulting in slave address 111011X0 ('X' is determined by state of SDO pin). Then the master sends pairs of register addresses and register data. The transaction is ended by a stop condition. This is depicted in Figure 7.



Figure 7: I²C multiple byte write (not auto-incremented)

5.2.2 I²C read

To be able to read registers, first the register address must be sent in write mode (slave address 111011X0). Then either a stop or a repeated start condition must be generated. After this the slave is addressed in read mode (RW = '1') at address 111011X1, after which the slave sends out data from auto-incremented register addresses until a NOACKM and stop condition occurs. This is depicted in Figure 8, where two bytes are read from register 0xF6 and 0xF7.



Figure 8: I²C multiple byte read

Überprüfung der Adresse per I2Cscan:

SD0 offen: I2C device found at address 0x76

SD0 an GND: I2C device found at address 0x76

SD0 an 3.3 V: I2C device found at address 0x77

Inbetriebnahme

I2C-Adresse 0x77 (SD0 an VCC)

Library: Adafruit BMP280

Sensor wird erkannt, Messung funktioniert